Zack Dalley
IS 6812-001

Business Problem Statement: Home Credit Default Risk

Business Situation

Home Credit is looking to help the unbanked population by providing a safe and positive borrowing experience. Many customers have limited credit histories, which makes standard credit scoring processes insufficient and leaves these customers to be taken advantage of by untrustworthy lenders. Thus, Home Credit is using alternative data to evaluate creditworthiness of these customers.

Business Problem

It's difficult to evaluate applicants with thin credit files because misclassifying creditworthy individuals leads to missed revenue and limited loan growth, while approving high-risk applicants increases loan default rates and credit losses. Currently, rigid rule-based models fail to fully leverage the alternative data, creating a gap between aggressive growth targets and effective risk management.

Expected Benefits

Expected benefits to the company include increasing total loan approval rates for creditworthy unbanked applicants by 8% to 12%, as well as reducing overall portfolio losses by 5% to 8% through improved precision on high-risk applicants. Additionally, we aim to reduce average underwriting processing time via automated risk scoring. All of this will increase revenue and decrease risk.

Success Metrics

Success will be measured against Home Credit's existing KPIs:

- Model Performance Metric: Area Under the ROC Curve. Looking for an AUC-ROC >= 0.78.
- Primary Financial Metric: Portfolio Loss Rate (Target: < 5.5%) and Net Interest Margin (Target improvement: +1.2%).

Analytics Approach

We will use gradient boosted decision trees trained on historical loan application data and other alternative data available to Home Credit. Customer-level predictions will output a probability score that will map to risk tiers for lending to them.

Scope

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In Scope:

- Data cleaning, feature engineering, modeling using Home Credit's historical and supplemental data.
- List of customers with predictions mapped to risk tiers.
- Executive presentation outlining model performance, risks, and projected financial ROI.

Project Execution & Milestones

- Phase 1: Business and Data Understanding (Weeks 1-2)
- Phase 2: Data preparation and Modeling (Weeks 3-5)
- Phase 3: Business Evaluation (Weeks 6-7)
- Phase 4: Deployment Strategy, List Deliverable, Final Presentation (Weeks 8-10)